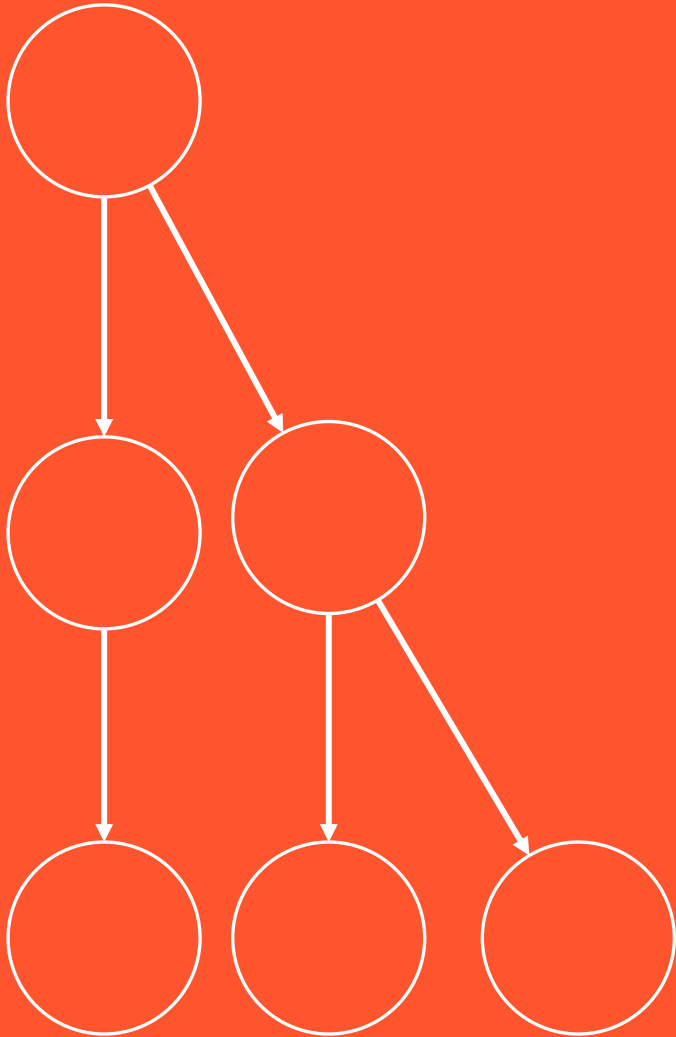




BCOG 100

Pattern Recognition

Lecture 7.3 — Pattern Recognition in the Cortex



UNIVERSITY OF
ILLINOIS
URBANA - CHAMPAIGN

To Start: Which Answer Would a Brain Use?

In your group, three minutes.

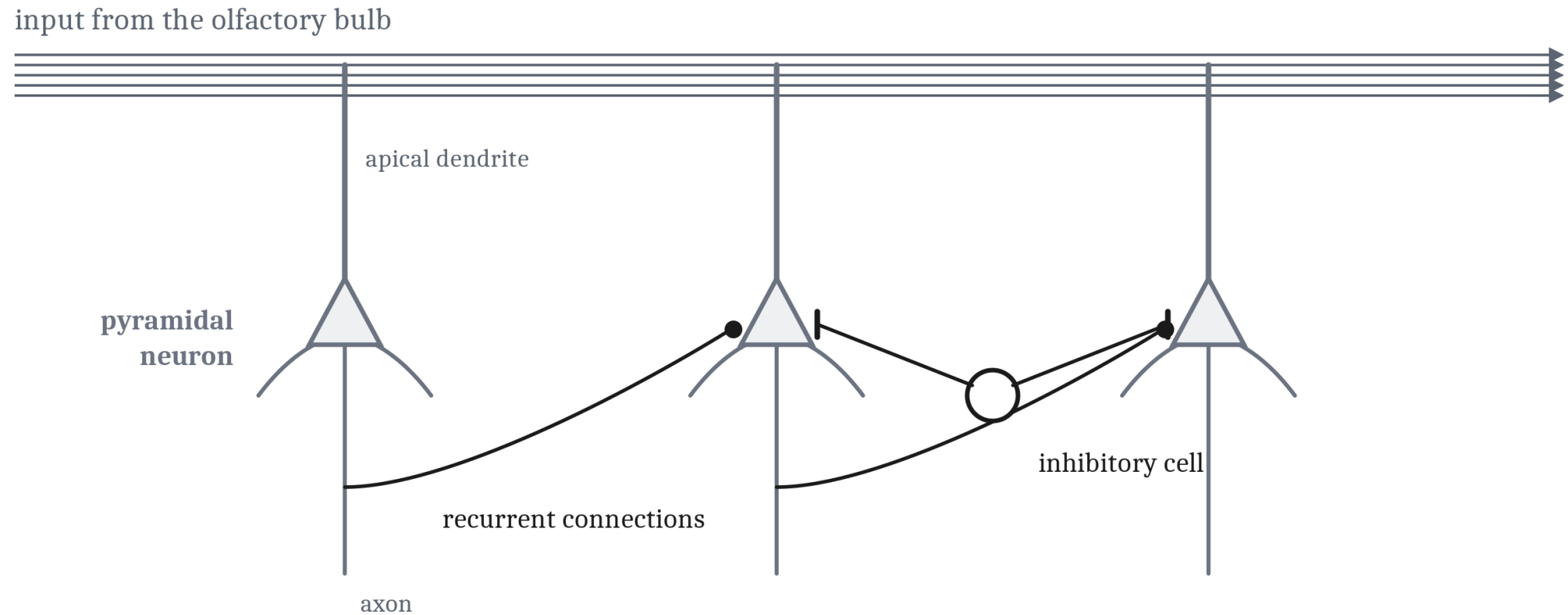
A hidden layer's weights can be set by passing errors back, left as random wiring with only the output trained, or set by competition with no teacher.

Which seems most plausible for a brain? What would we look for in brain tissue to tell the three apart?

The Oldest Cortex

The **piriform cortex**: three layers, input straight from the olfactory bulb with no thalamic relay, built around the **pyramidal neuron**.

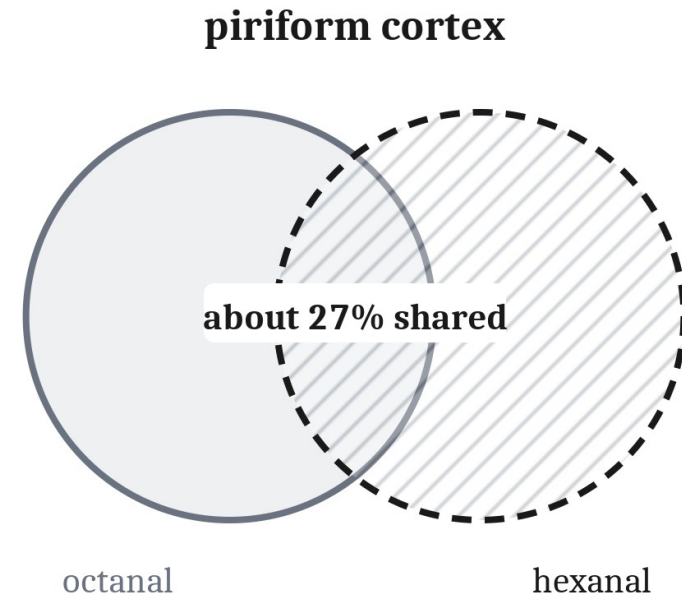
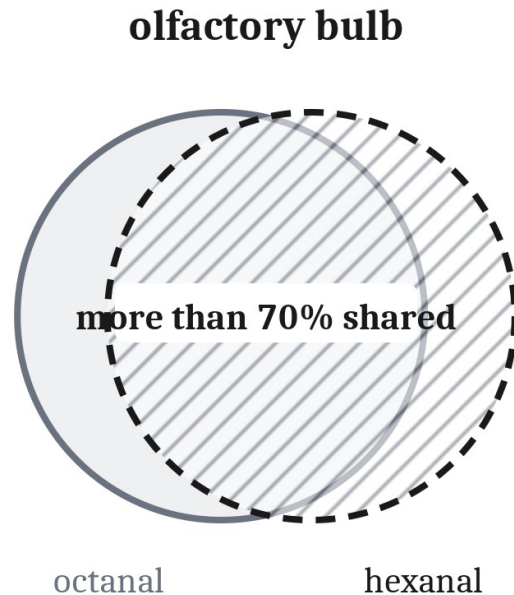
The basic circuit of the piriform cortex



Pulling Similar Smells Apart

Pattern separation: making overlapping inputs less alike, so a later stage can tell them apart. Measured here; random wiring is a model that fits it.

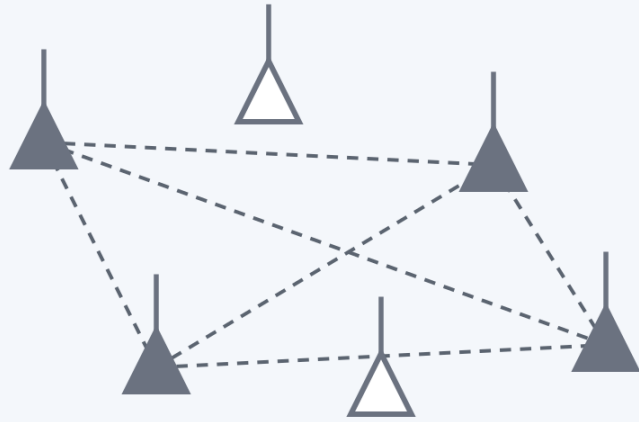
Two similar odors, at two stages



Completing a Smell from Part of It

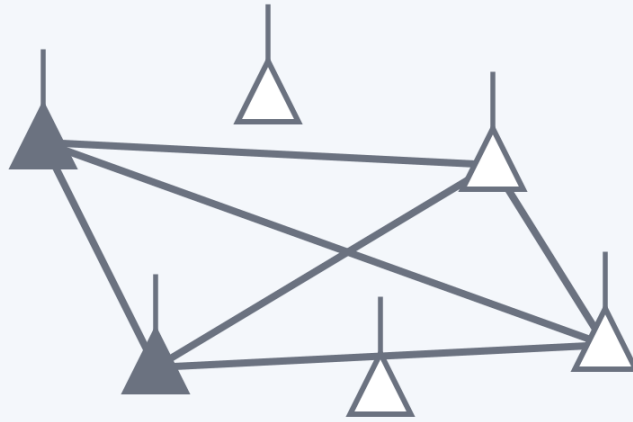
Auto-association stores a pattern in a circuit's own connections; **pattern completion** recovers it from part. In the piriform cortex, mostly a model result.

1. A smell is stored



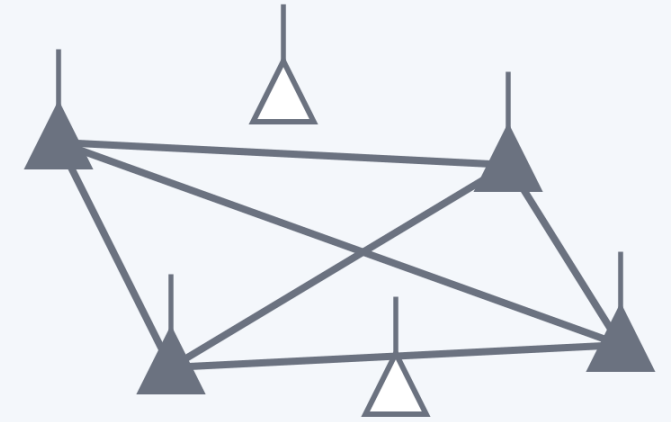
Hebb's rule strengthens the connections among the active cells

2. Later: only part of it



a faint trace, or a partial mixture, activates two of them

3. The rest is completed



the strengthened connections activate the other two

Auto-association stores the pattern in the circuit's own connections. Pattern completion recovers it from part.

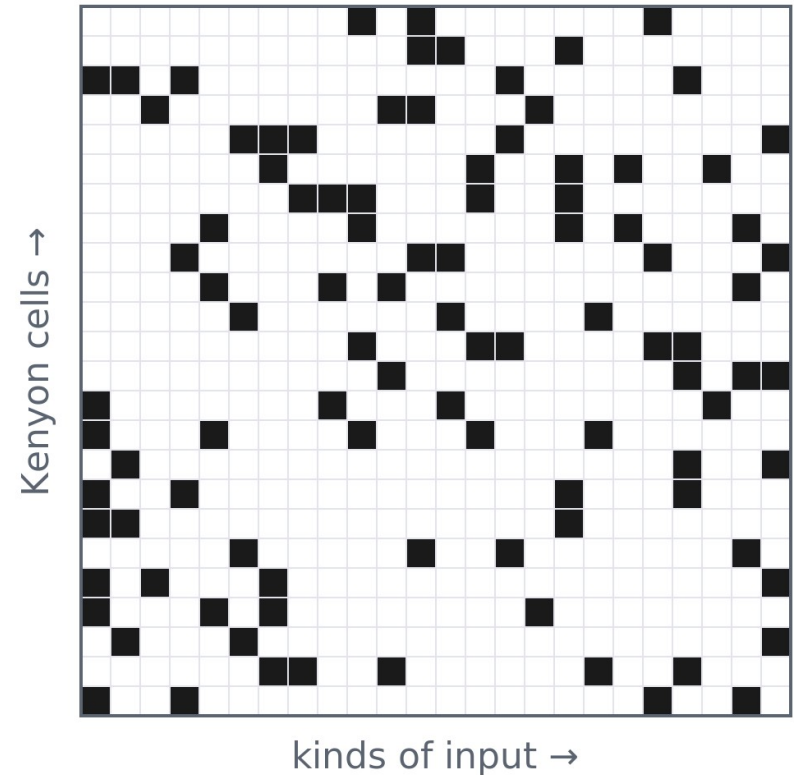
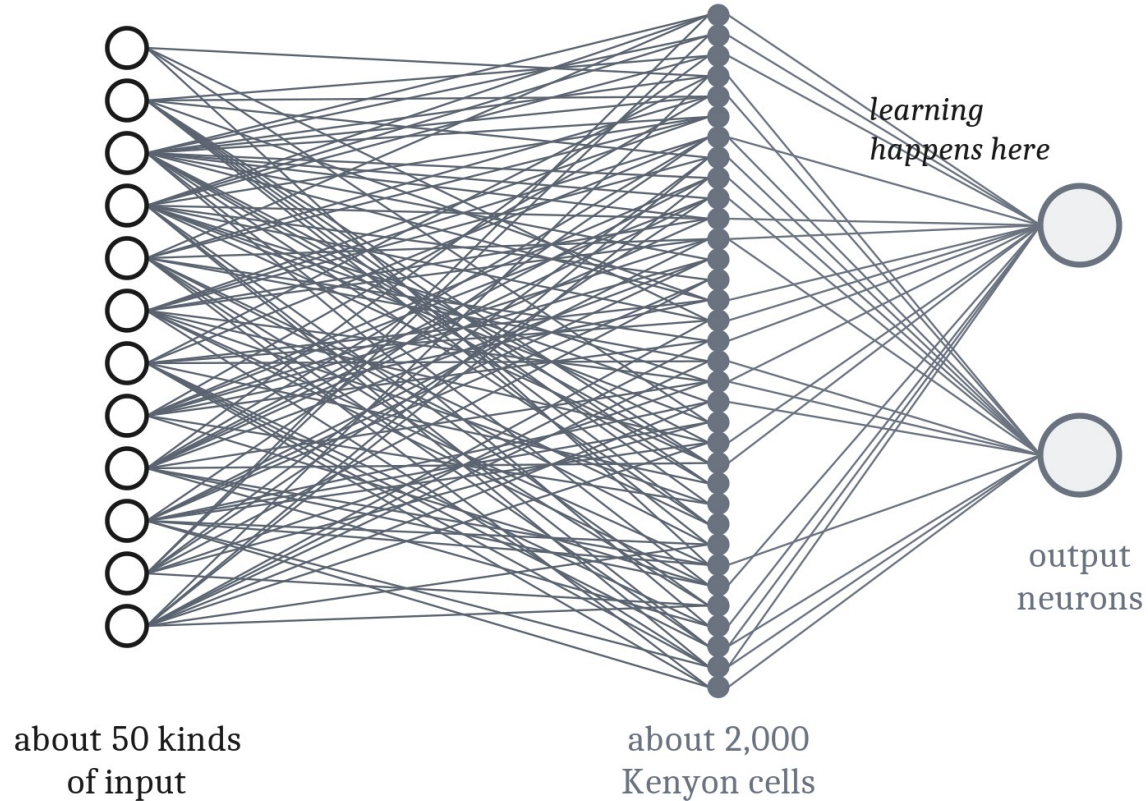
In the piriform cortex this is mostly a model result: direct tests in animals are few.

Filled: an active pyramidal neuron. Schematic, not a recording.

A Fly, a Bird, and a Fish

The fly's **mushroom body**: about 50 kinds of input spread at random across about 2,000 cells, never trained. Learning happens downstream.

Random expansion in the fruit fly's mushroom body



One Plan, Many Areas

Module 6's uniform sheet, corrected: not one circuit copied identically, but one circuit plan, repeated with variations.

One plan, repeated with variations

an area that receives the senses

layer 1
layer 2/3
layer 4
layer 5
layer 6

layer 4: prominent

Same cell classes, much the same wiring; counts and strengths differ.

the motor cortex

layer 1
layer 2/3
layer 5
layer 6

layer 4: nearly absent

Where a cell's input comes from

excitatory synapses, input layer of V1

from the thalamus: under a tenth

Even in the layer that receives the thalamus, most input comes from other cortical cells.

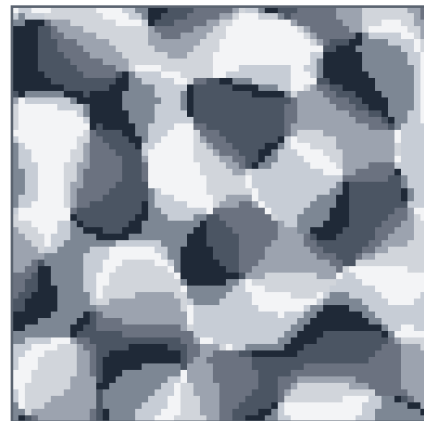
Does one plan mean one computation? Still open.

Areas resemble one another the way hands resemble feet. Layer thicknesses are schematic.

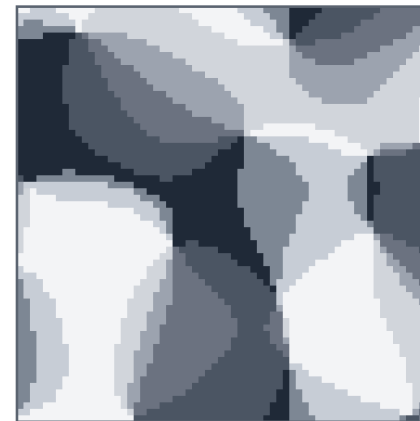
The Rewired Ferrets

Input shapes what a cortical area does. Not that areas are interchangeable, and not that the ferrets saw normally.

The rewired ferrets



visual cortex



rewired auditory cortex

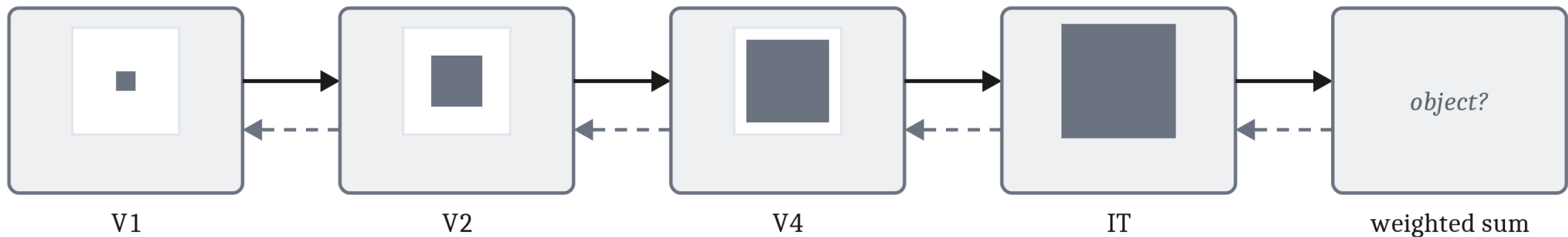
Orientation maps drawn to show the comparison, not traced from data. Each shade is one preferred orientation.

The Visual Cortex Beside a Network

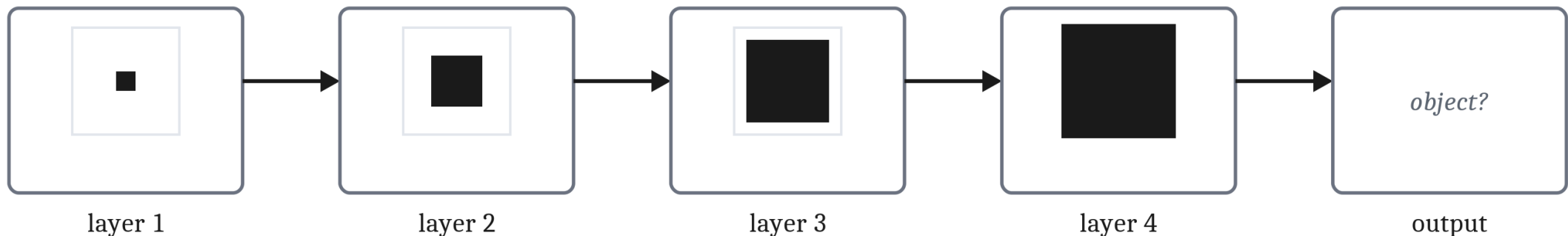
Complex cells built from **simple cells**, as Hubel and Wiesel tentatively proposed, would be convolution then pooling. Up the **ventral stream**, cells gain **tolerance**, never full invariance.

The ventral stream beside a convolutional network

ventral stream



convolutional network

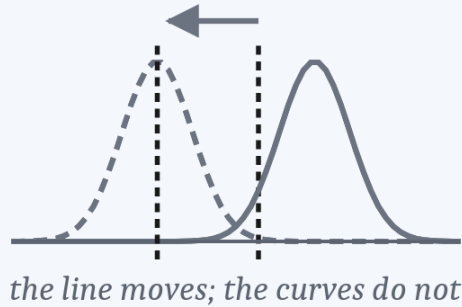


Receptive fields (the dark squares) grow at each stage. Dashed arrows: feedback, present only in the cortex.

Criterion and Sensitivity, in Tissue

The criterion lives in how excitable a neuron is, which neuromodulators change. Sensitivity lives in the representation built upstream.

The criterion: excitability

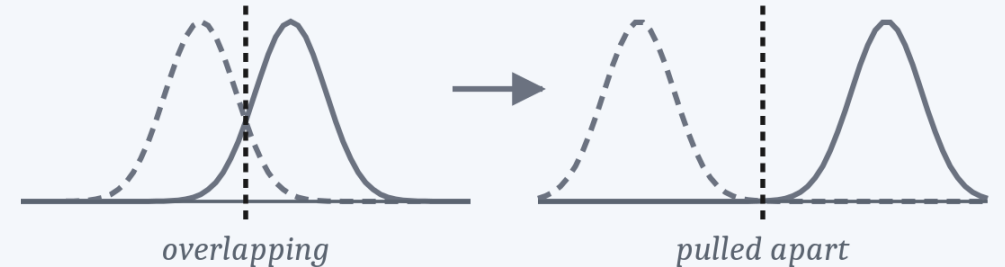


How close a neuron's resting state sits to firing.

Neuromodulators change it.

The hungry and the fed worm of Module 4: same input, same wiring, different chemistry, a moved criterion.

Sensitivity: the representation upstream



How well the input a neuron receives

separates the things that matter:

piriform separation, mushroom-body expansion,
the stages of the ventral stream.

Signal detection's two quantities, with two physical homes.

Could the Cortex Pass Errors Back?

What the pallium lacks is an external teacher, not necessarily an error. How an error would be delivered is still open.

Literal backpropagation would need three things no one has found in the cortex

Matching weights

feedback connections carrying the same weights as the forward connections they run beside

Signed errors

error signals positive or negative, over a huge range of values

Silent feedback

feedback that changes synapses without changing the activity it passes through

No direct evidence for anything like it in the cortex, and none that rules it out.

Learning from before and after

Feedback changes a neuron's activity, and it learns from the difference. Can approximate backpropagation with signals a neuron could have.

Predictive coding

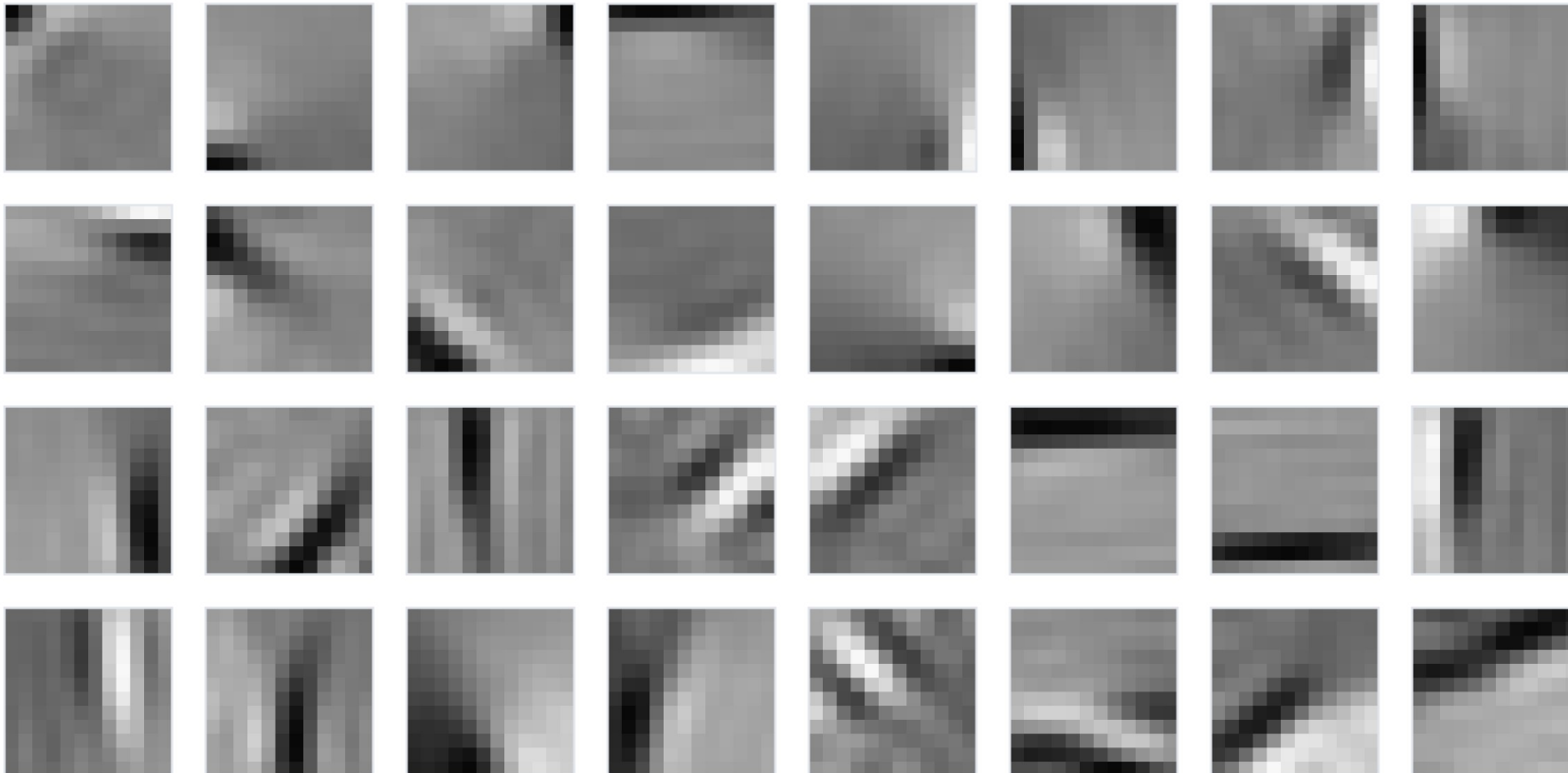
Feedback carries predictions down; forward connections carry what they missed. Its error neurons lack clear support.

Both are open hypotheses.

Few Active at Once

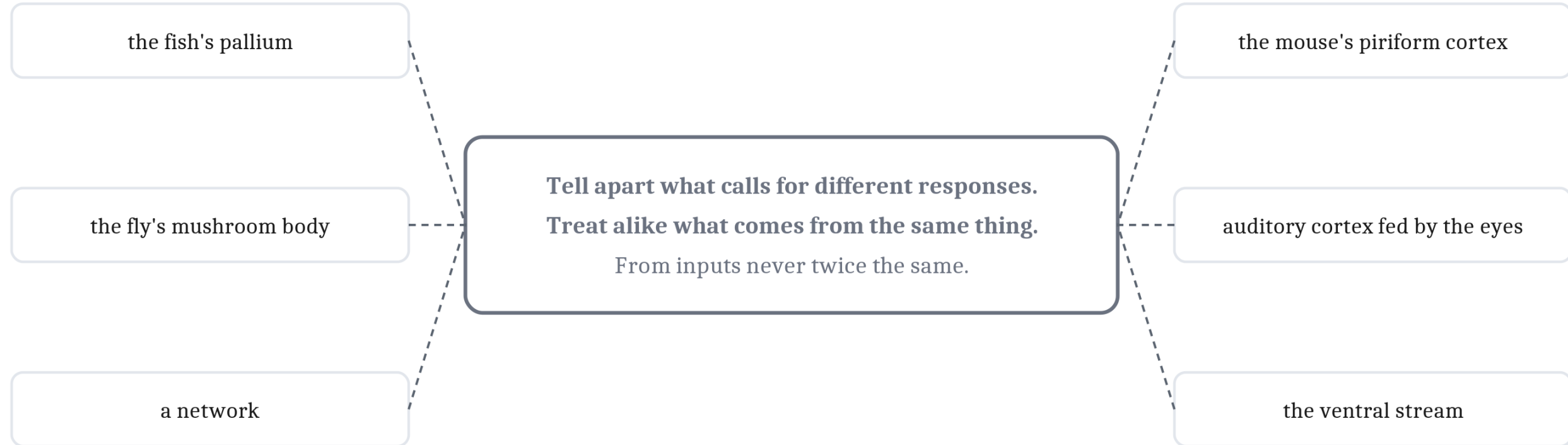
The energy budget limits how many neurons can be active at once, and a code that sparse also suits recognition.

Receptive fields learned without a teacher



One Plan, Many Uses

A structure that does the same thing to every input is doing something very general: it re-represents whatever reaches it, until a single straight line can do the rest.



The computational level holds steady while the implementation is rebuilt beneath it.

Recognition says what a thing is, not why it is there or what acting on it would do.

Reading Braille with the Visual Cortex

In your group, four minutes.

In people blind from birth or early childhood, the visual cortex becomes active when they read Braille by touch. Briefly disrupting that area with magnetic stimulation disrupts their Braille reading.

Using the rewired ferrets, what do these findings suggest? And what do they not show?

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